



IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

Mohammad Zeeshan
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Outline

- Executive Summary
- Introduction
- Methodology
- Results
- Conclusion
- Appendix

Executive Summary

- Summary of methodologies

- Data collection via SpaceX REST API
- Data collection with Web Scraping from Wikipedia using BeautifulSoup
- Data wrangling
- Exploratory Data Analysis with SQL over SQLite
- Exploratory Data Analysis with Visualization
- Interactive Visual Analytics with Folium/Dash
- Machine Learning Prediction and Validation

- Summary of all results

- Success Rate over Time: An improvement in success rate was observed over time.
- Success Rate by Launch Site: KSC LC-39A shows the highest success rate of all launch sites.
- Success Rate by Orbit: ES-L1, SSO, HEO, and GEO were observed to have the highest success rates.
- Payload: Heavier payloads have a high failure rate early on with a significant improvement over time.
- Predictive Analysis: The DecisionTreeClassifier algorithm has proven highly accurate in predicting landing outcomes.

Introduction

- SpaceX advertises Falcon 9 rocket launches on its website with a cost of 62 million dollars.
- Other providers cost upward of 165 million dollars each.
- Much of the savings is due to SpaceX's ability to reuse the first stage.
- If we can determine if the first stage will land, we can determine the cost of a launch.

Introduction

In this capstone project, we aim to predict the successful landing of the Falcon 9 first stage. SpaceX advertises rocket launches at a significantly lower cost compared to other providers, largely due to their ability to reuse the first stage of the rocket. By accurately predicting landing success, we can estimate launch costs and provide valuable insights for companies bidding against SpaceX.

The project endeavors to...

- Collect as much publicly available data as possible via the various channels like API and Websites.
- Wrangle the data to improve its quality and prepare it for analysis.
- Explore the processed data with SQLite and various visualizations.
- Drill down into finer levels of detail by splitting the data into groups defined by categorical variables.
- Build, evaluate, and refine predictive models for discovering additional insights.

Problem Statement

- Predict the successful landing of SpaceX's Falcon 9 first stage, a critical factor in determining the cost-efficiency of reusable rocket launches.

Why should we address this problem?

- Solving this problem allows companies to make competitive bids against SpaceX by accurately estimating launch costs.

Questions

- What is the historical success rate of Falcon 9 first stage landings?
- What factors most significantly influence the success or failure of a landing?
- Can we develop a predictive model that accurately forecasts the outcome of a landing?

Section 1

Methodology

Methodology

Executive Summary

- **Data collection methodology**
 - Retrieval and consolidation from multiple [SpaceX API](#) endpoints
 - Web scraping tabular data from [Wikipedia](#)
- **Perform data wrangling**
 - Extracted relevant records
 - Target Labeling
 - Flattened fields and resolved missing values
 - Feature selection and pipeline integration
- **Perform exploratory data analysis (EDA) using visualization and SQL**
 - Visualize variable relationships
 - Look at the data in aggregate
 - Boundary and performance identification
- **Perform interactive visual analytics using Folium and Plotly Dash**
 - Mark all launch sites on a map
 - Mark successful and failed launches
 - Calculate distances to proximate locations
 - Provide for interactive exploration of the data
- **Perform predictive analysis using classification models**
 - Build, evaluate, and compare several predictive classification models

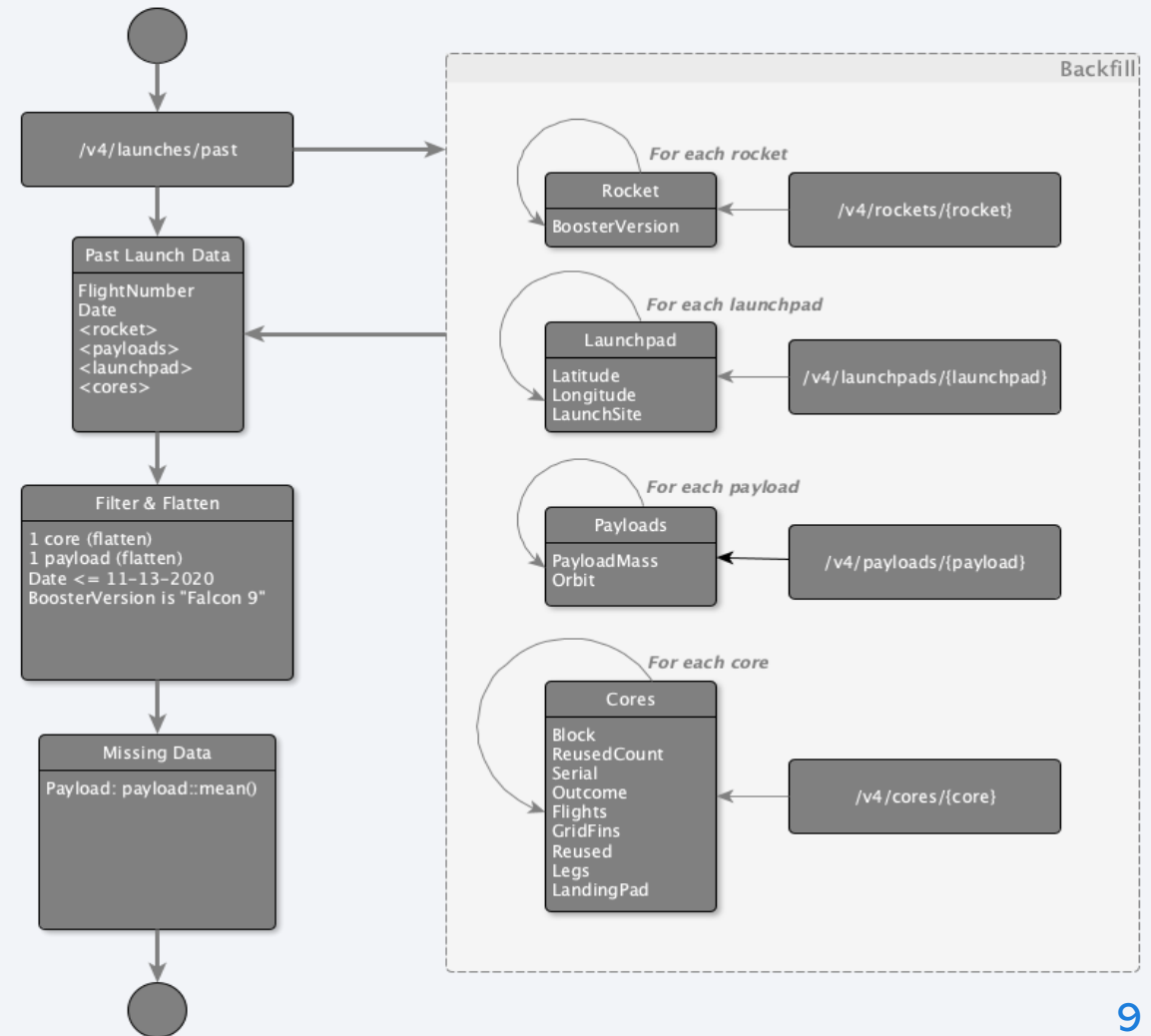
Data Collection

Data was collected using a combination of retrieval techniques:

- **SpaceX REST API Integration:** Programmatically retrieved structured JSON telemetry records directly from past launch endpoints ('/v4/launches/', 'v4/rockets/', 'v4/payloads/', 'v4/cores/'), isolating primary booster configurations and initial core specifications.
- **Automated Web Scraping:** Extracted historical launch tables from [Wikipedia](#) using BeautifulSoup , using clean text processing to remove superscript footnote markers and parse alphanumeric strings.
- **Feature Ingestion:** Built a consolidated baseline dataset capturing critical tracking variables including Flight Number, Date, Booster Serial, Payload Mass, Destination Orbit, and specific Landing Pad locations.

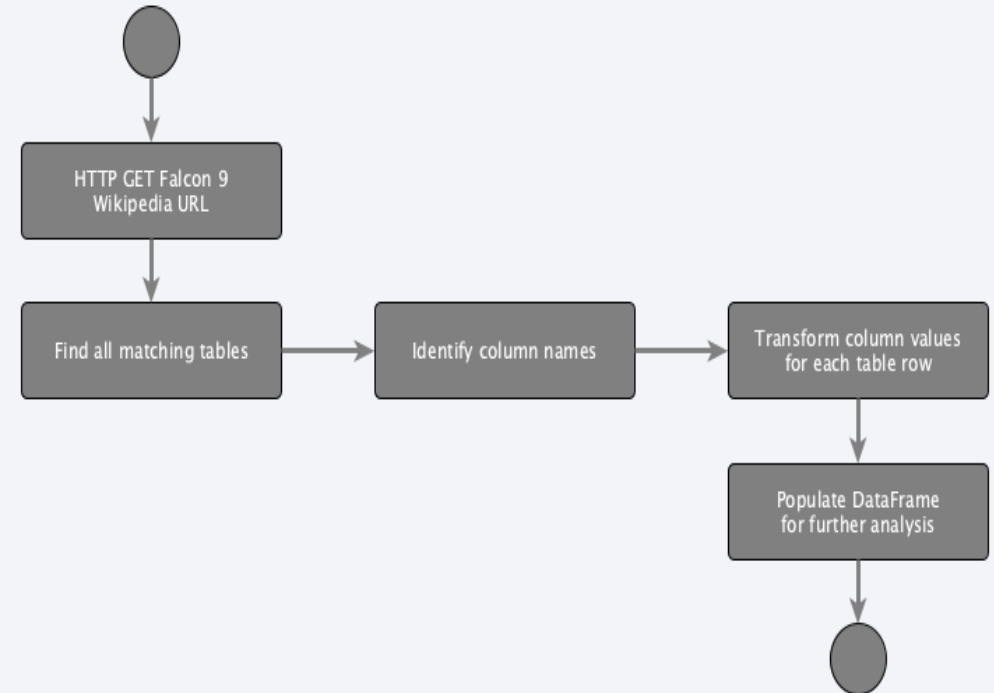
Data Collection – SpaceX API

- **REST API Querying:** Programmatically fetched historical telemetry and launch metadata using Python's 'requests' library to target the official SpaceX endpoint ('/v4/launches/past').
- **Dataset Normalization & Filtering:** Filtered out early test vehicles (Falcon 1) to establish a baseline data matrix focused entirely on Falcon 9 orbital flights.
- **Relational Feature Unfolding:** Unpacked linked sub-JSON IDs by systematically calling supplementary endpoints to resolve critical attributes for core configurations, booster serial codes, specific payload data, and landpad coordinates.
- **Tabular Structuring:** Converted the combined API dictionary outputs into a structured Pandas DataFrame containing core tracking columns (Flight Number, Date, Mass, Orbit, Site, Outcome).
- **Notebook (GitHub URL)**



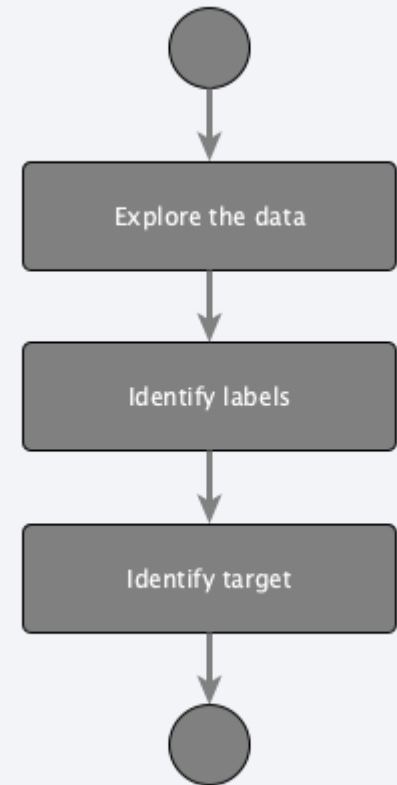
Data Collection - Scraping

- Web scraping workflow:
 - **Target Extraction:** Used Python's 'BeautifulSoup' library to parse the raw HTML structure of the Wikipedia article titled `"List of Falcon 9 and Falcon Heavy launches"`.
 - **HTML Element Parsing:** Filtered and navigated the document object model (DOM) tree to locate and isolate structural 'wikitable' data containers.
 - **String Cleaning Workflow:** Applied regular expressions to systematically strip out citation brackets, newline markers, and footnote superscript artifacts.
 - **Tabular Structuring:** Converted the extracted, cleaned text rows into an organized Pandas DataFrame mapped to clear, descriptive column headers.
- Notebook (GitHub URL)



Data Wrangling

- **Goals:**
 - Use Exploratory Data Analysis (EDA) to find patterns/trends in the data.
 - Determine the labels for training supervised models.
- **Steps:**
 - Find missing values as a percentage of each attribute
 - Identify column types, numerical or categorical
 - Show launches per Launch Site
 - Show distribution of Orbit type in the data set
 - Explore outcomes, and group them by binary outcome (success or failure)
 - Store outcome as "Class" label, to be used as target value in training
- Notebook (GitHub URL)



Exploratory Data Analysis (EDA) via Visualization

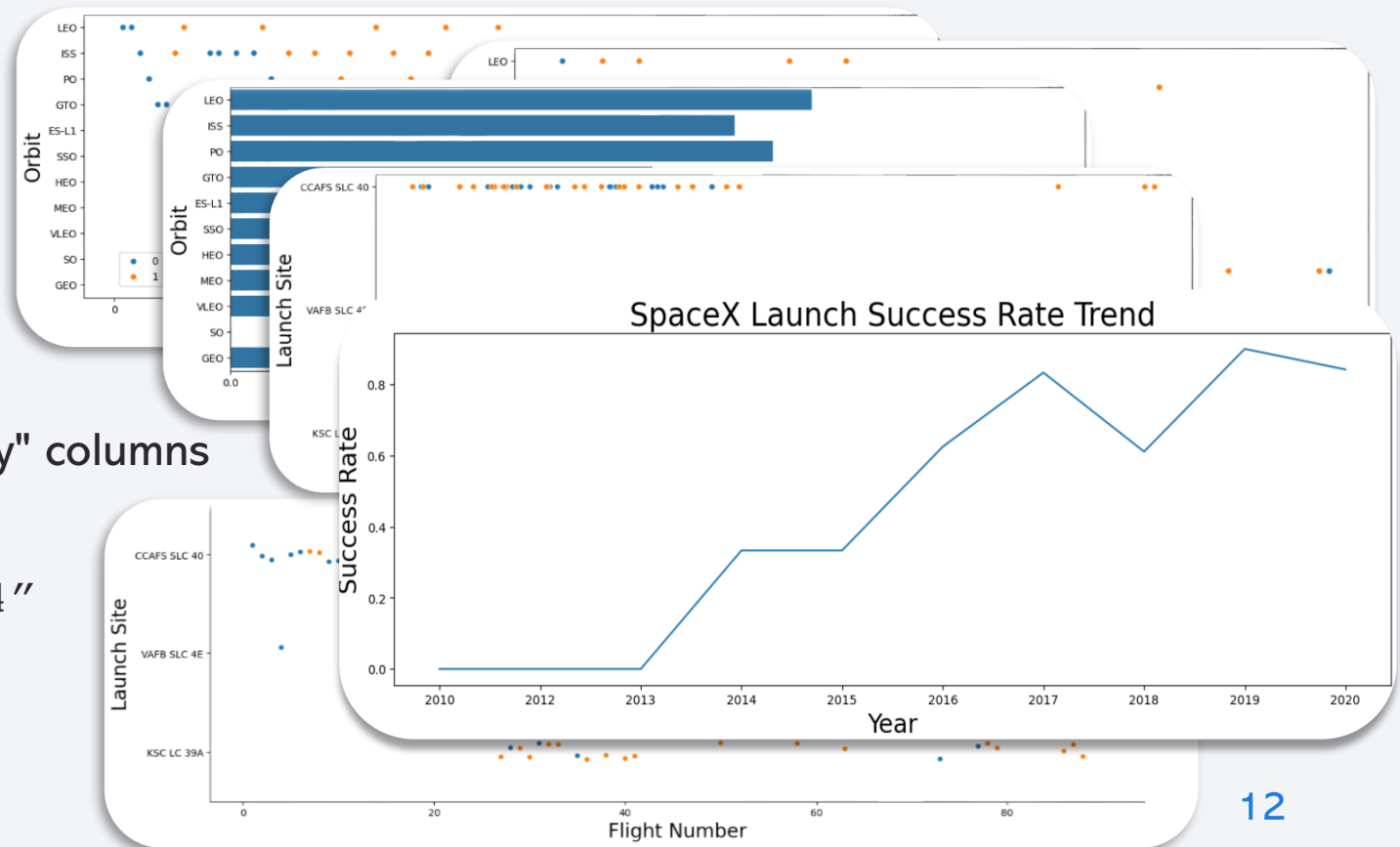
- Visualize relationships to gain insight into the importance of each variable:

- Flight Number and Outcome
- Flight Number and Launch Site
- Payload and Launch Site
- Orbit and Outcome
- Flight Number and Orbit
- Payload and Orbit
- Yearly success rate

- Expand categorical variables into "dummy" columns using the `get_dummies()` method

- Keep all numerical columns as `float64`

- Notebook (GitHub URL)

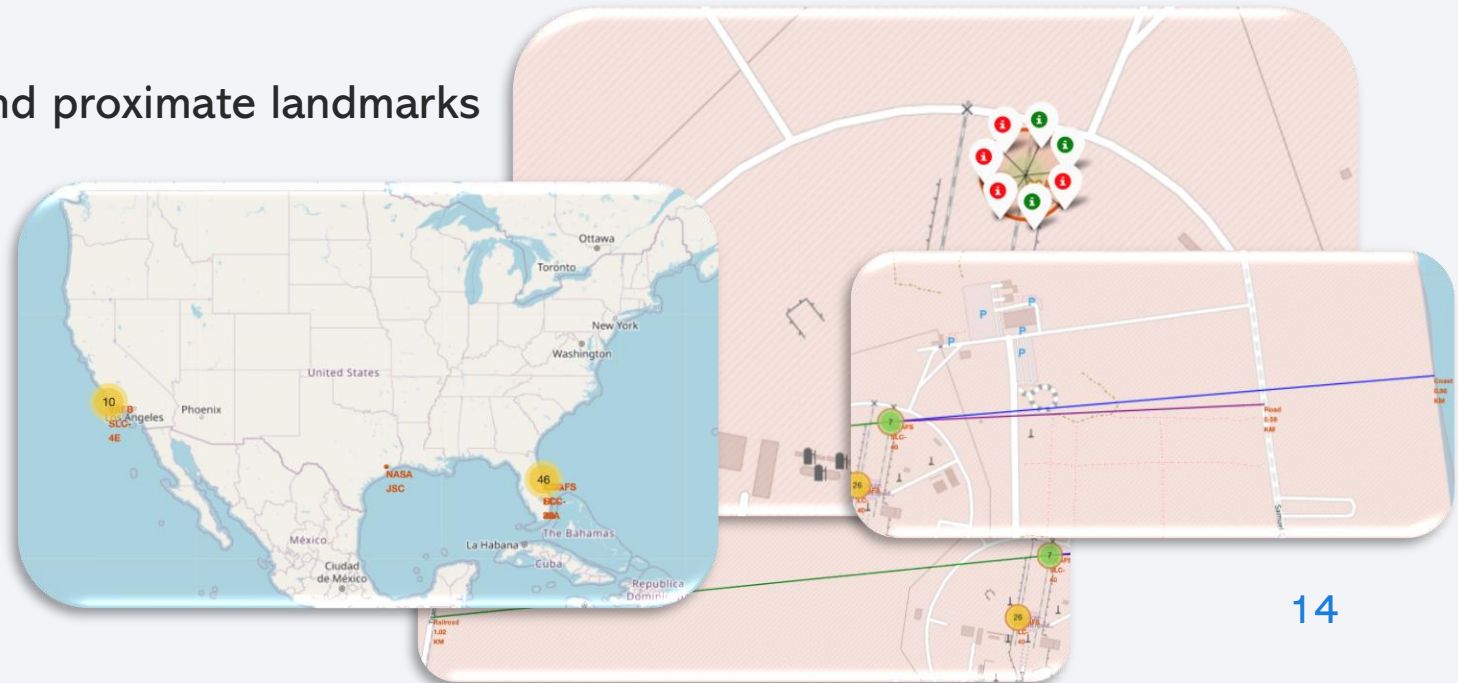


EDA with SQL (SQL)

- SQL Queries Performed:
 - Show each unique launch site
 - Show 5 records where launch site names begin with 'CCA'
 - Display the total payload mass carried by boosters launched by 'NASA (CRS)'
 - Display the average payload mass carried by the v1.1 Falcon 9 booster
 - List the date of the first successful ground landing outcome
 - List the booster versions with successful outcomes landing on the drone ship with payloads between 4000kg and 6000kg.
 - List the number of successful and failed mission outcomes
 - List all of the booster versions that carried the max payload mass
 - List the month name, outcome, booster version, and launch site for missions with failure outcomes landing on a drone ship in 2015.
 - Show the distribution of outcomes between June 4th, 2010 and March 20th, 2017
- Notebook (GitHub URL)

Build an Interactive Map with Folium

- To find geographical patterns in the data the following items were marked on a map of launch sites:
 - All Launch Sites
 - Successful and Failed Launches
 - Distances between a launch site and proximate landmarks
- Notebook (GitHub URL)



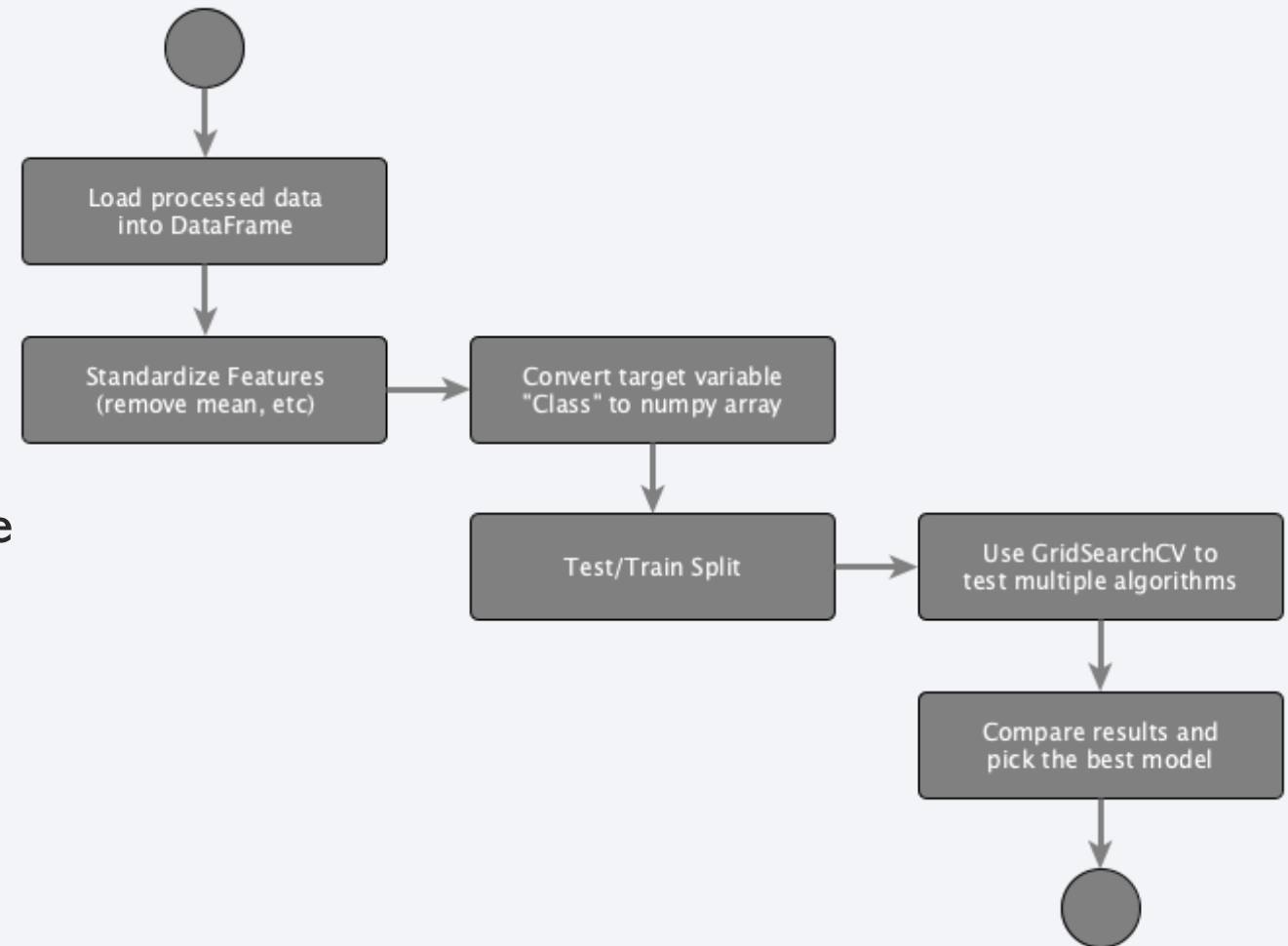
Build a Dashboard with Plotly Dash

To enable interactive exploration of the data, a Plotly Dash dashboard was developed to include:

- A dropdown selector to choose a Launch Site, affecting:
 - A pie chart
 - All sites selected: shows the breakdown of successful outcomes across all sites
 - A launch site selected: shows the breakdown of successful vs failed launches for the given site
 - A scatter plot
 - All sites selected: shows outcome by payload mass and booster version for all sites
 - A launch site selected: shows outcome by payload mass and booster version for the given site
- A Payload Mass range selector that filters data points on the scatter plot
- Plotly Dash App (GitHub URL)

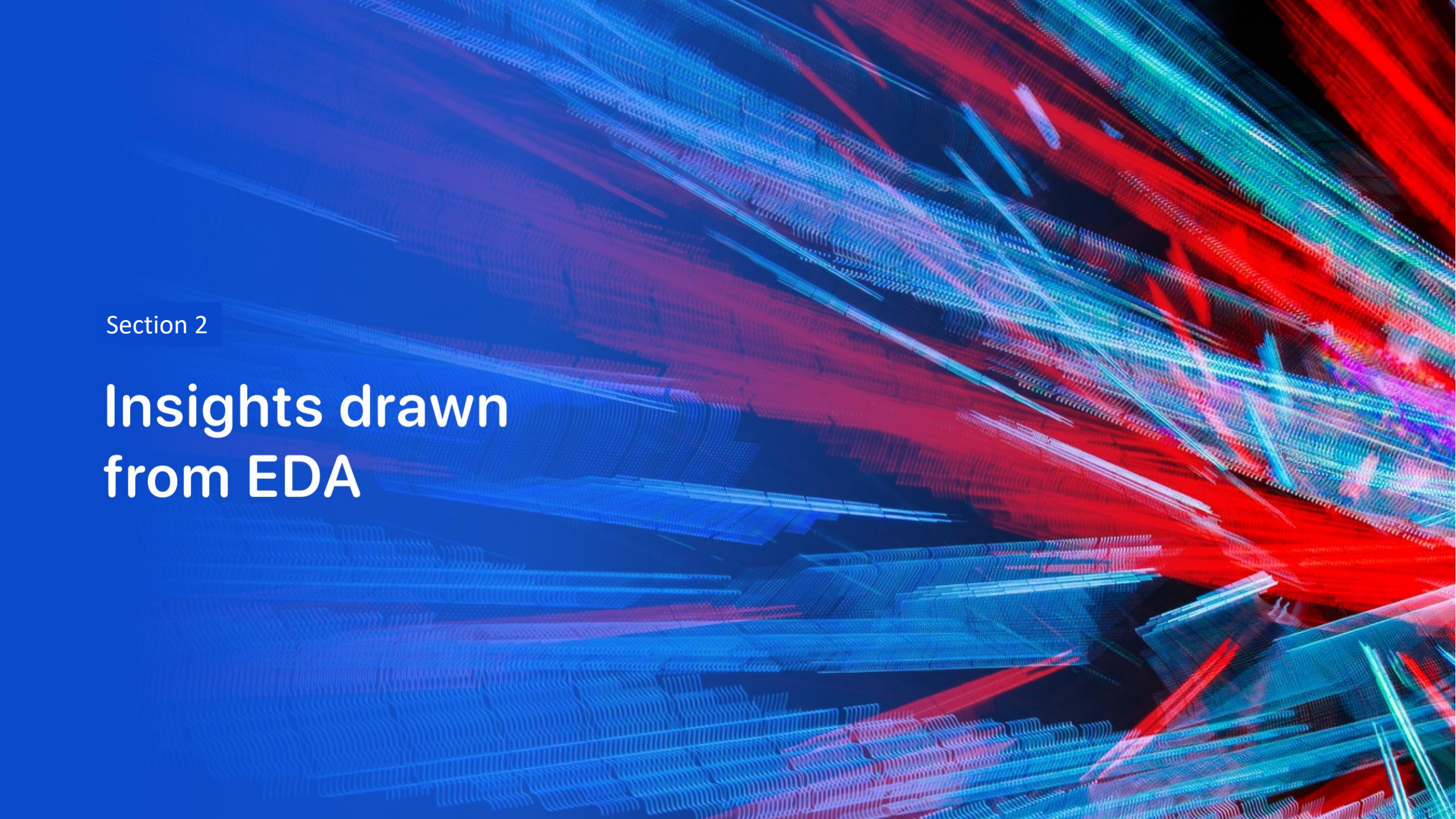
Predictive Analysis (Classification)

- Load data
- Apply StandardizedScaler on X
- Convert Y to numpy array
- Split training and testing data
- Use GridSearchCV to test hyperparameters for multiple algorithms:
 - Logistic Regression
 - SVC
 - Decision Tree Classifier
 - K Neighbors Classifier
- Notebook (GitHub URL)



Results

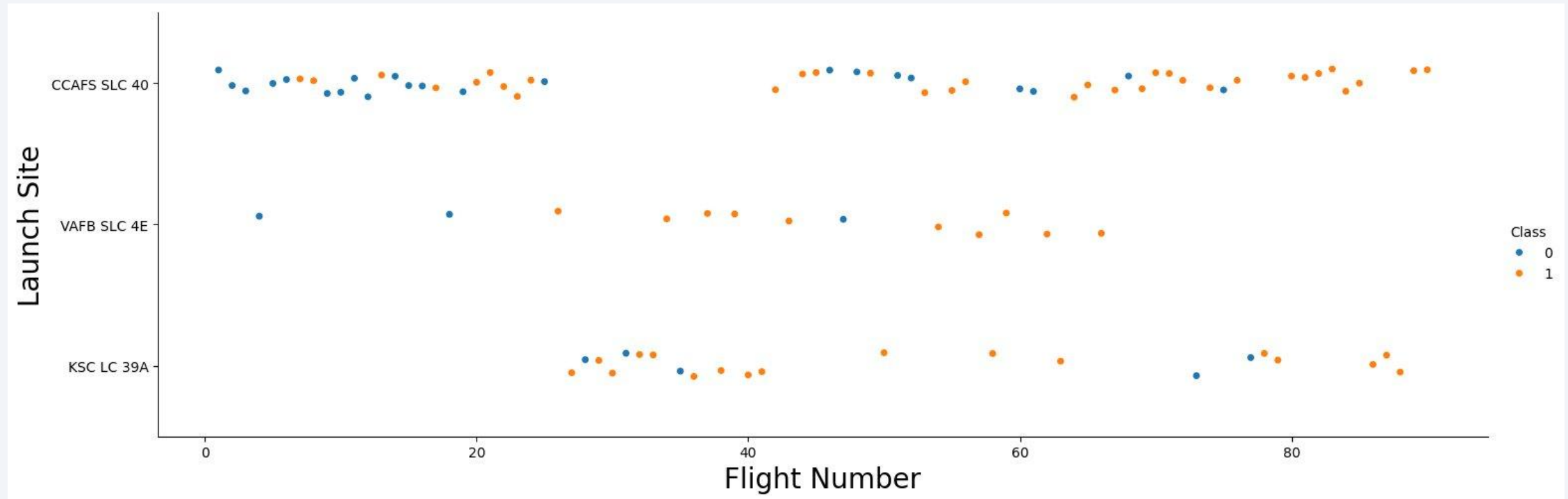
- Exploratory data analysis results
- Interactive analytics demo in screenshots
- Predictive analysis results

The background of the slide is an abstract composition. It features a dark blue base color. Overlaid on this are numerous diagonal streaks in shades of red and cyan. A faint, light blue grid pattern is also visible, particularly in the lower half of the image. The overall effect is dynamic and technological.

Section 2

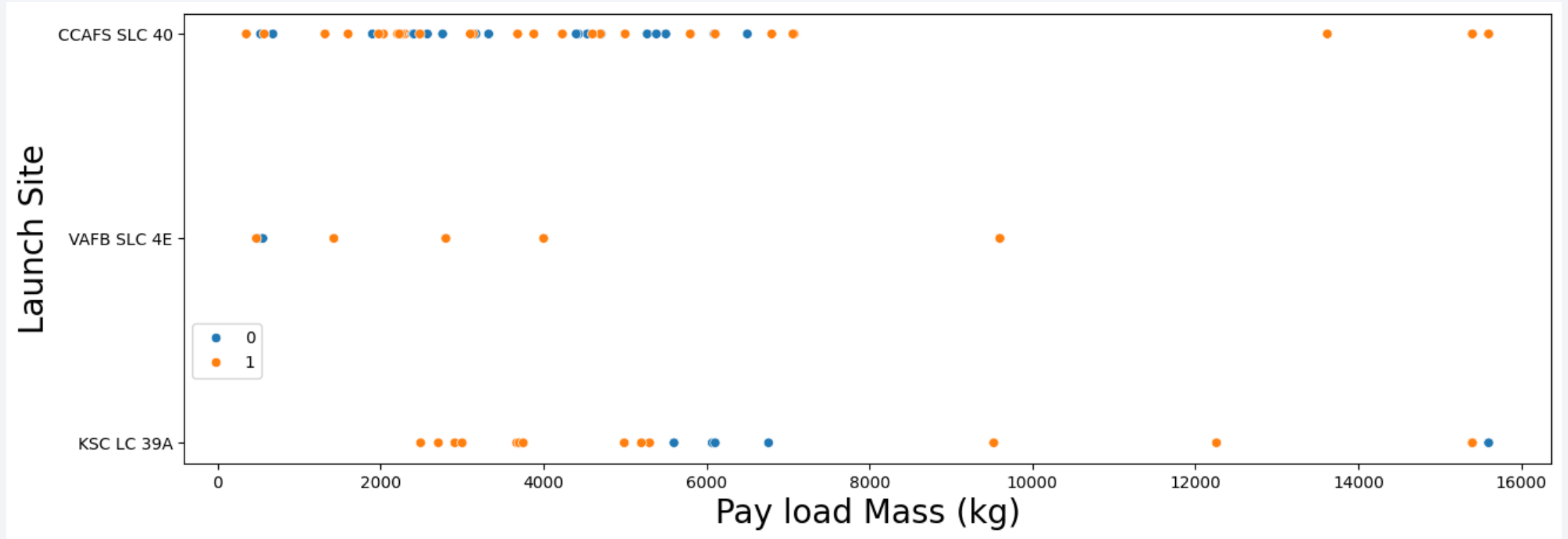
Insights drawn from EDA

Flight Number vs. Launch Site



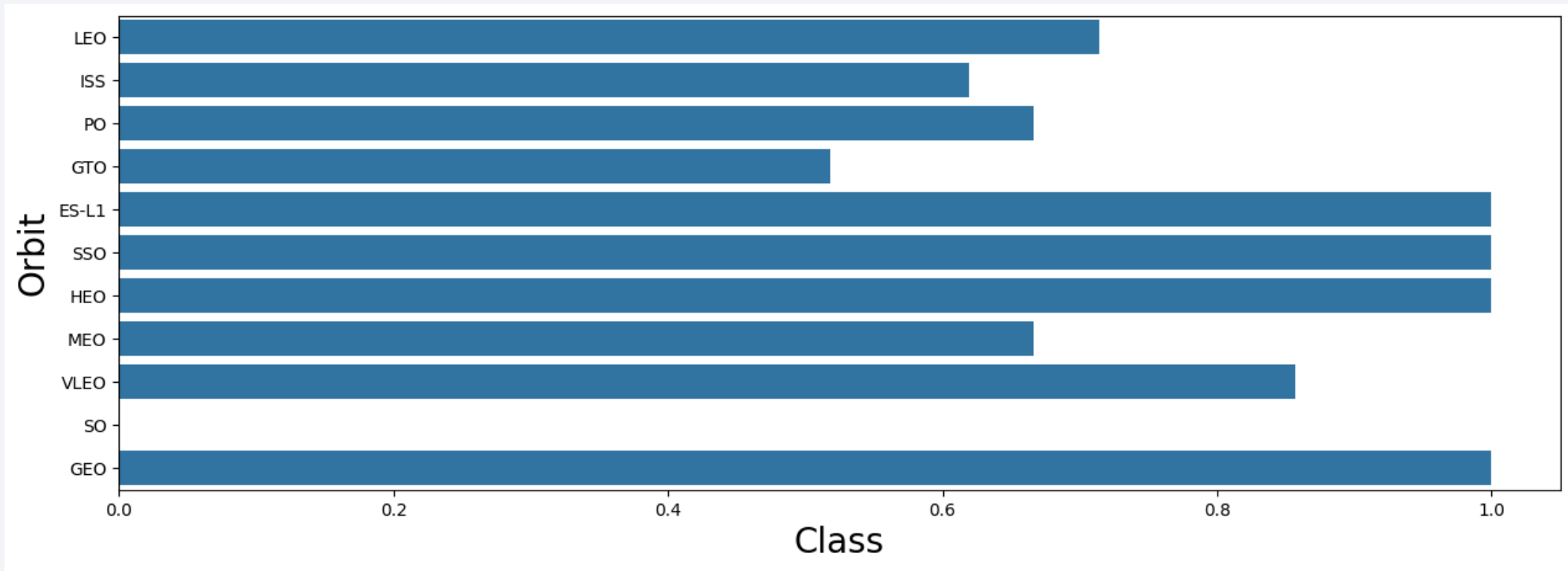
- All sites show a mix of first stage landing successes and failures, with successes increasing over time.
- Early flights predominantly resulted in failures, indicating improvements to technology or process.
- While CCAFS SLC 40 has the most total flights, VAFB SLC 4E appears to have a relatively higher proportion of successful landing outcomes.

Payload vs. Launch Site



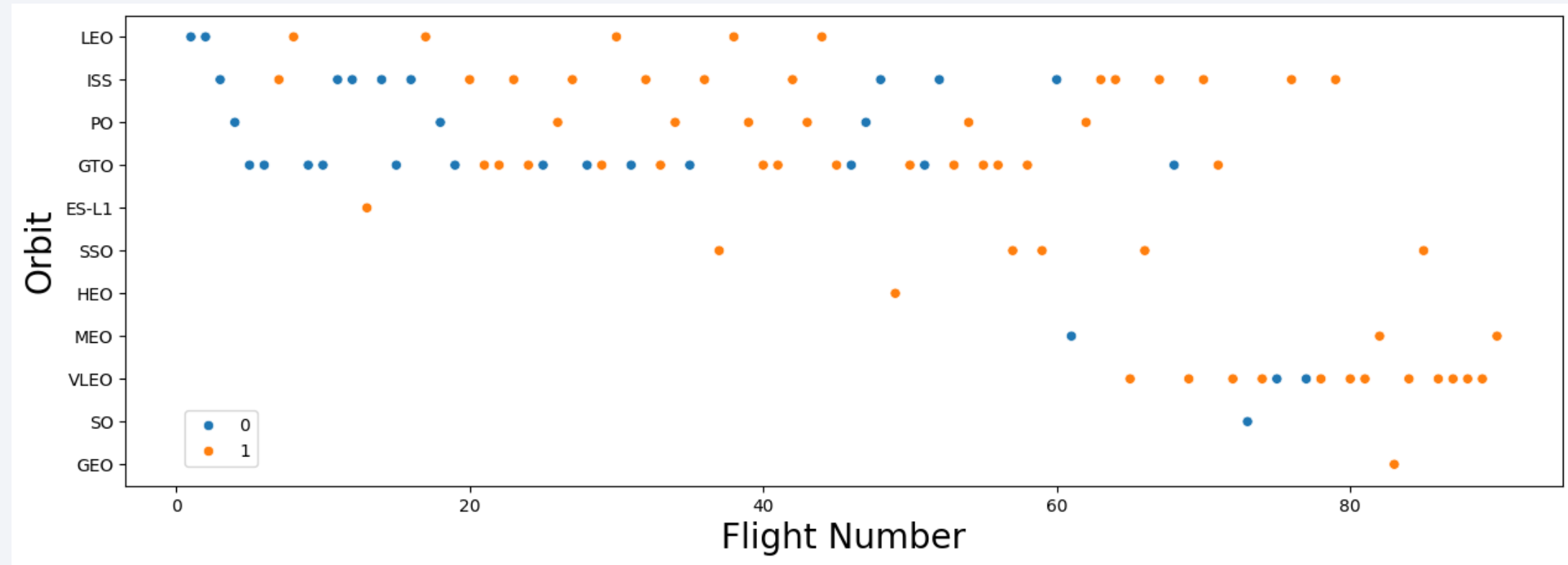
- All sites show a wide range of payload weights, from light to heavy payloads.
- Early flights trend toward lighter payloads, representing the bulk of the landing failures.
- This suggests technological or operational improvements lead to a greater rate of success with heavier payloads.

Success Rate vs. Orbit Type



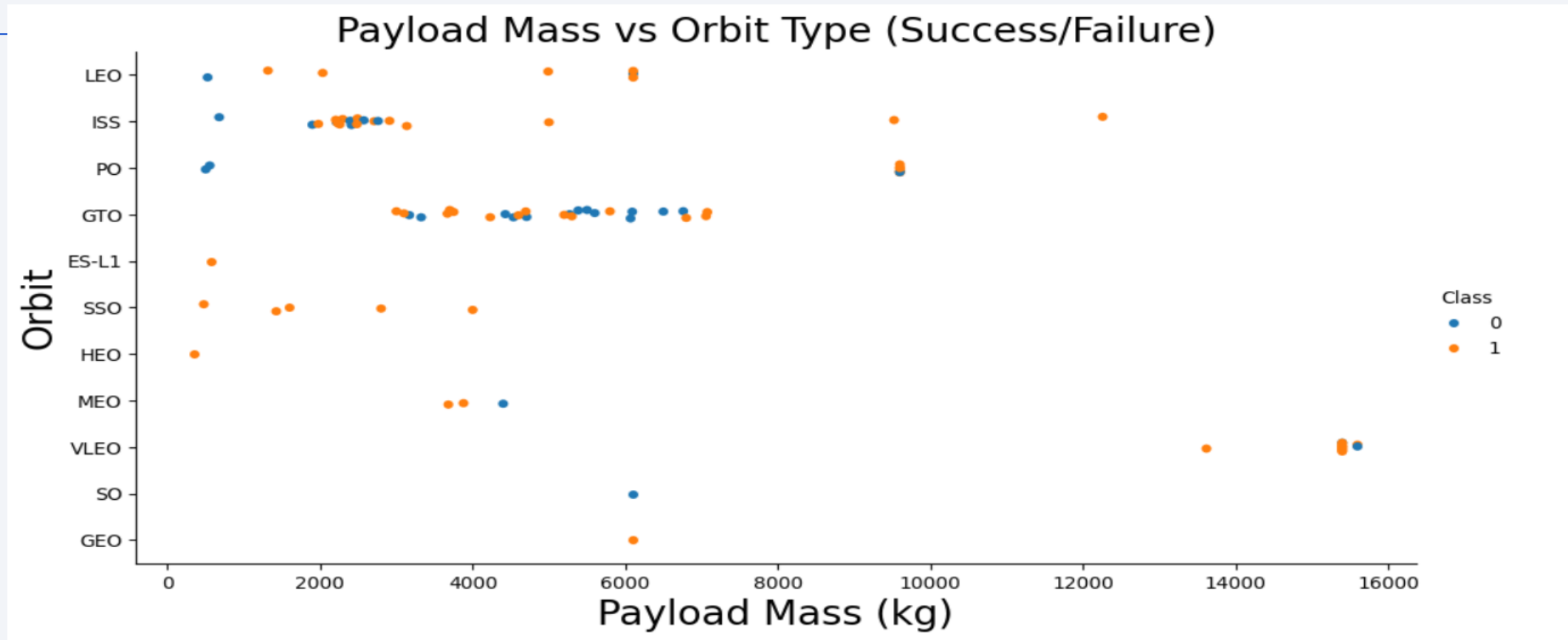
- Some orbits, such as ES-L1, SSO, HEO, and GEO consistently show high success rates.
- Others such as GTO show more mixed outcomes, suggesting some orbit types may introduce operational or technological challenges.
- With only one launch, there is not enough data for the SO orbit type to provide an accurate analysis.

Flight Number vs. Orbit Type



- Many orbits are represented throughout the flight number range, some orbits are not attempted until later flights.
- There is a noticeable improvement in landing success as flight numbers increase, indicating an accumulation of experience and ongoing improvements.

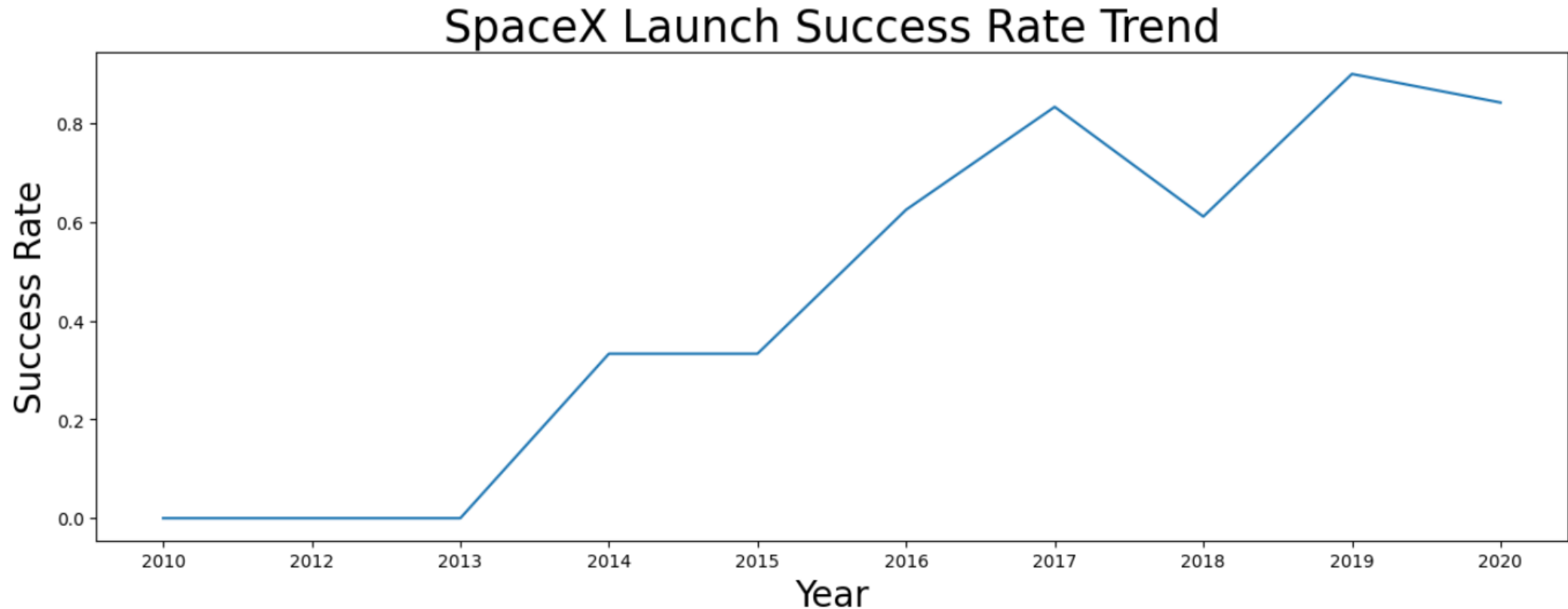
Payload vs. Orbit Type



- Many orbits are represented across a wide range of payload masses, but others like SSO, MEO, HEO and GEO show a generally lower range.
- Orbits with a constrained payload range, tend to show a higher rate of landing success.
- While payload mass does not appear to directly determine mission success, its interplay with orbit suggests a significant correlation.

Launch Success Yearly Trend

- Yearly trend shows a consistent progression from early challenges to high reliability in first stage landings over time.
- From 2016 onward, SpaceX experienced year over year improvement in success rate with a minor setback in 2018.



All Launch Site Names

There are four unique Launch Sites

- CCAFS LC-40
- VAFB SLC-4E
- KSC LC-39A
- CCAFS SLC-40

```
SELECT DISTINCT Launch_Site from SPACEXTABLE;
```

Launch Site Names Begin with 'CCA'

- First five records where launch sites begin with `CCA`

Date	Time (UTC)	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS__KG_	Orbit	Customer	Mission_Outcome	Landing_Outcome
2010-06-04	18:45:00	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecraft Qualification Unit	0	LEO	SpaceX	Success	Failure (parachute)
2010-12-08	15:43:00	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1, two CubeSats, barrel of Brouere cheese	0	LEO (ISS)	NASA (COTS) NRO	Success	Failure (parachute)
2012-05-22	7:44:00	F9 v1.0 B0005	CCAFS LC-40	Dragon demo flight C2	525	LEO (ISS)	NASA (COTS)	Success	No attempt
2012-10-08	0:35:00	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1	500	LEO (ISS)	NASA (CRS)	Success	No attempt
2013-03-01	15:10:00	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2	677	LEO (ISS)	NASA (CRS)	Success	No attempt

```
SELECT * FROM SPACEXTABLE WHERE Launch_Site LIKE 'CCA%' LIMIT 5;
```

Total Payload Mass

The total payload carried by boosters from NASA (CRS) is **45,596kg**.

```
SELECT SUM(PAYLOAD_MASS__KG_) AS TOTAL_PAYLOAD  
FROM SPACEXTABLE WHERE Customer = 'NASA (CRS)';
```

Average Payload Mass by F9 v1.1

The average payload mass carried by booster version F9 v1.1 is **2,534.67kg**.

```
SELECT AVG(PAYLOAD_MASS__KG_) AS AVG_PAYLOAD_MASS  
FROM SPACEXTABLE WHERE Booster_Version LIKE 'F9 v1.1%';
```

First Successful Ground Landing Date

The first successful landing outcome on ground pad occurred on **December 22nd, 2015**.

```
SELECT MIN(Date) as LaunchDate
FROM SPACEXTABLE
WHERE Landing_Outcome = 'Success (ground pad)';
```

Successful Drone Ship Landing with Payload between 4000 and 6000

The boosters which have successfully landed on drone ship and had payload mass greater than 4000 but less than 6000 are:

Booster	Payload Mass
F9 FT B1022	4,696kg
F9 FT B1026	4,600kg
F9 FT B1021.2	5,300kg
F9 FT B1031.2	5,200kg

```
SELECT Booster_Version, PAYLOAD_MASS__KG_  
FROM SPACEXTABLE  
WHERE Landing_Outcome = 'Success (drone ship)'  
AND PAYLOAD_MASS__KG_ BETWEEN 4000 AND 6000;
```


Total Number of Successful and Failure Mission Outcomes

- Total number of successful and failure mission outcomes

Mission Status	Count
Failure	1
Success	100

```
SELECT CASE
    WHEN Mission_Outcome LIKE 'Success%' THEN 'Success'
    WHEN Mission_Outcome LIKE 'Failure%' THEN 'Failure'
END as Mission_Status, COUNT(*)
FROM SPACEXTABLE
GROUP BY Mission_Status;
```

Boosters Carried Maximum Payload

- The maximum payload sent was **15,600kg**.
- The boosters that carried the maximum payload are:

```
SELECT
  DISTINCT Booster_Version,
  PAYLOAD_MASS__KG_
FROM SPACEXTABLE
  WHERE PAYLOAD_MASS__KG_ = (
    SELECT MAX(PAYLOAD_MASS__KG_) FROM
    SPACEXTABLE
  )
ORDER BY Booster_Version;
```

Booster Version
F9 B5 B1048.4
F9 B5 B1048.5
F9 B5 B1049.4
F9 B5 B1049.5
F9 B5 B1049.7
F9 B5 B1051.3
F9 B5 B1051.4
F9 B5 B1051.6
F9 B5 B1056.4
F9 B5 B1058.3
F9 B5 B1060.2
F9 B5 B1060.3

2015 Launch Records

- List of failed landing outcomes in drone ship, their booster versions, and launch site names for in year 2015:

Month	Outcome	Booster	Launch Site
January	Failure (drone ship)	F9 v1.1 B1012	CCAFS LC-40
April	Failure (drone ship)	F9 v1.1 B1015	CCAFS LC-40

```
SELECT
CASE strftime('%m', Date)
WHEN '01' THEN 'January'
WHEN '02' THEN 'February'
WHEN '03' THEN 'March'
WHEN '04' THEN 'April'
WHEN '05' THEN 'May'
WHEN '06' THEN 'June'
WHEN '07' THEN 'July'
WHEN '08' THEN 'August'
WHEN '09' THEN 'September'
WHEN '10' THEN 'October'
WHEN '11' THEN 'November'
WHEN '12' THEN 'December'
END as Month,
Landing_Outcome, Booster_Version, Launch_Site, Date
FROM SPACEXTABLE
WHERE strftime('%Y', Date) = '2015' AND Landing_Outcome = 'Failure (drone ship)';
```

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

- Rank the count of landing outcomes between the date 2010-06-04 and 2017-03-20, in descending order:

```
SELECT Landing_Outcome, COUNT(*) as Count
FROM SPACEXTABLE
WHERE Date BETWEEN '2010-06-04' AND '2017-03-20'
GROUP BY Landing_Outcome
ORDER BY Count DESC;
```

Landing Outcome	Count
No attempt	10
Success (drone ship)	5
Failure (drone ship)	5
Success (ground pad)	3
Controlled (ocean)	3
Uncontrolled (ocean)	2
Failure (parachute)	2
Precluded (drone ship)	1

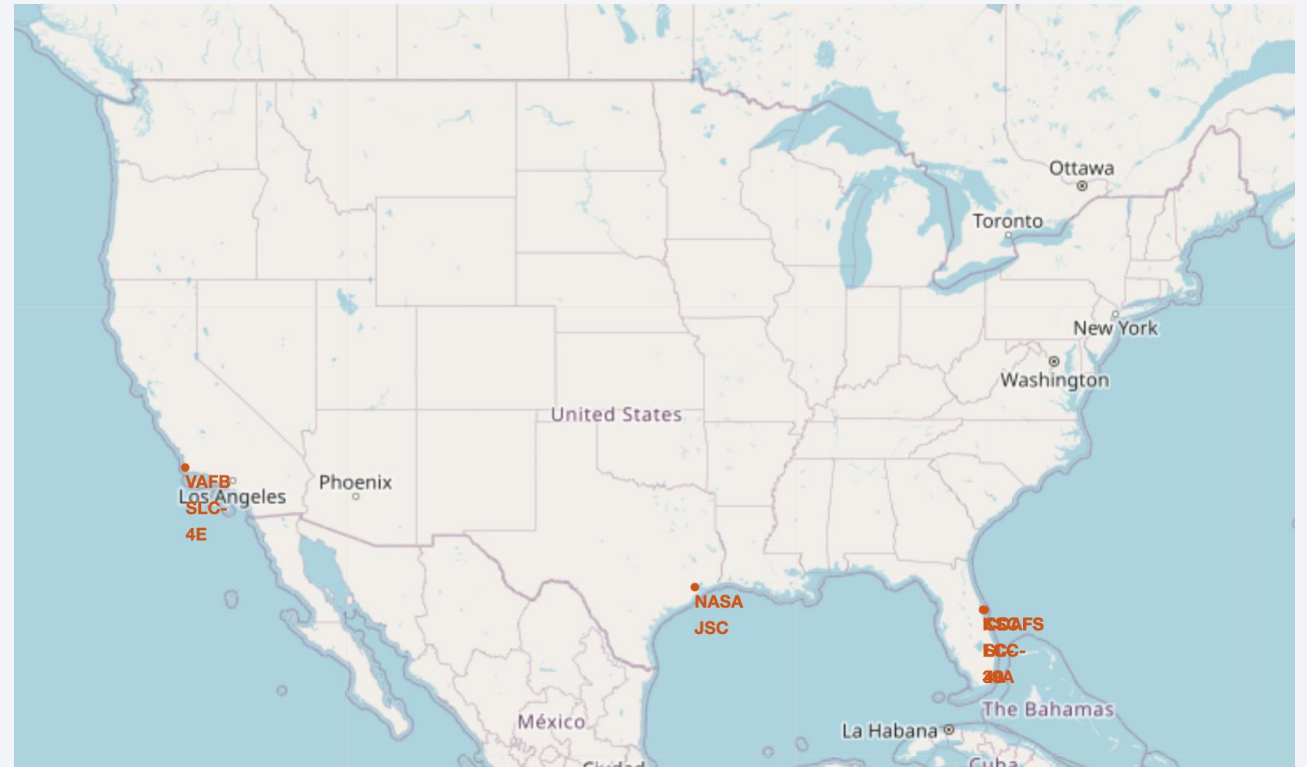
A satellite view of Earth from space, showing the curvature of the planet and city lights at night. The background is a deep blue gradient.

Section 3

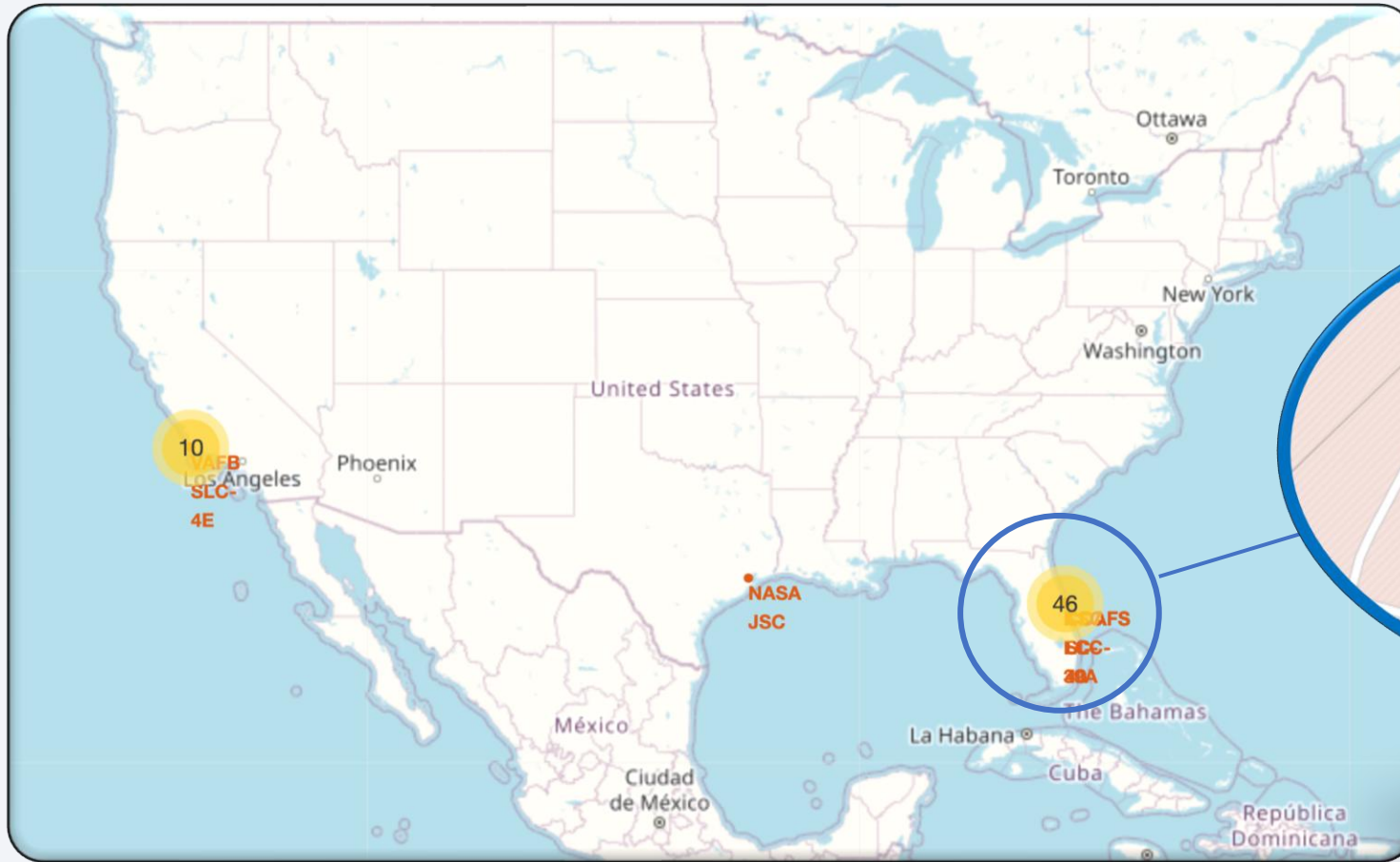
Launch Sites Proximities Analysis

Launch Site Locations

- Launch sites are located near coastal regions in Florida and California to reduce risk of catastrophic failures affecting human activities.

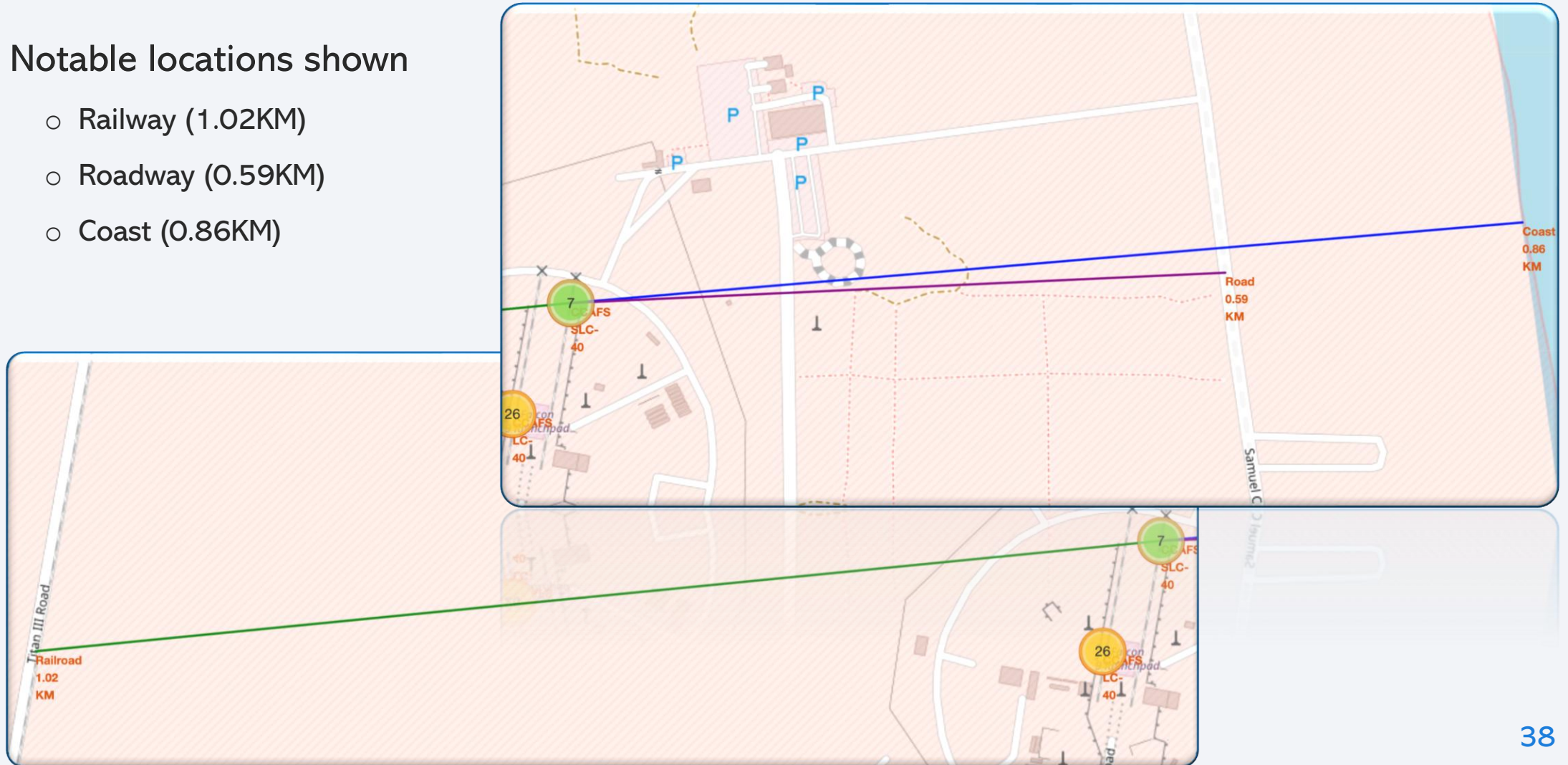


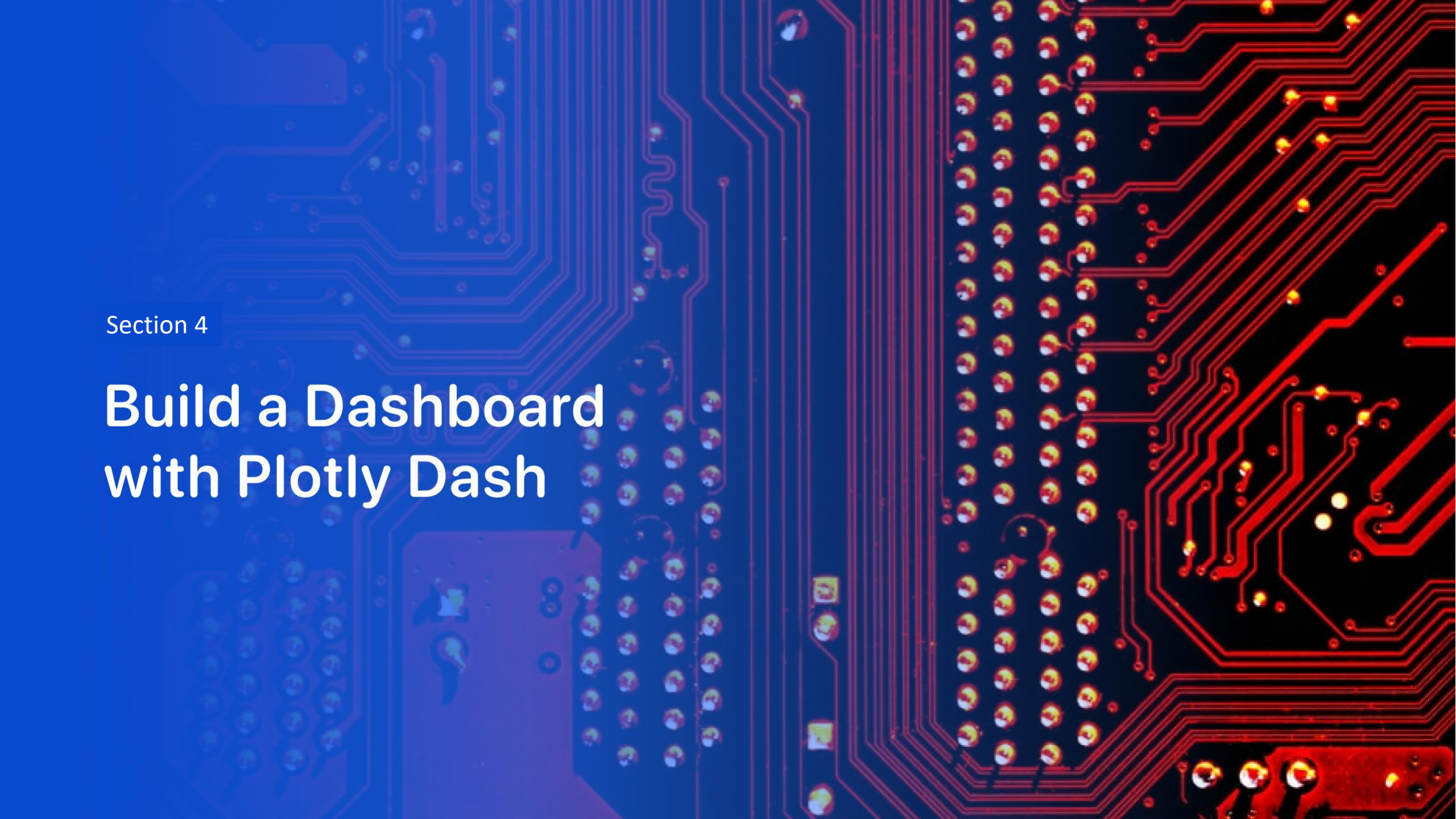
Launch Outcomes



Notable Proximate Locations

- Notable locations shown
 - Railway (1.02KM)
 - Roadway (0.59KM)
 - Coast (0.86KM)





Section 4

Build a Dashboard with Plotly Dash

All Launch Sites: Successful Landings

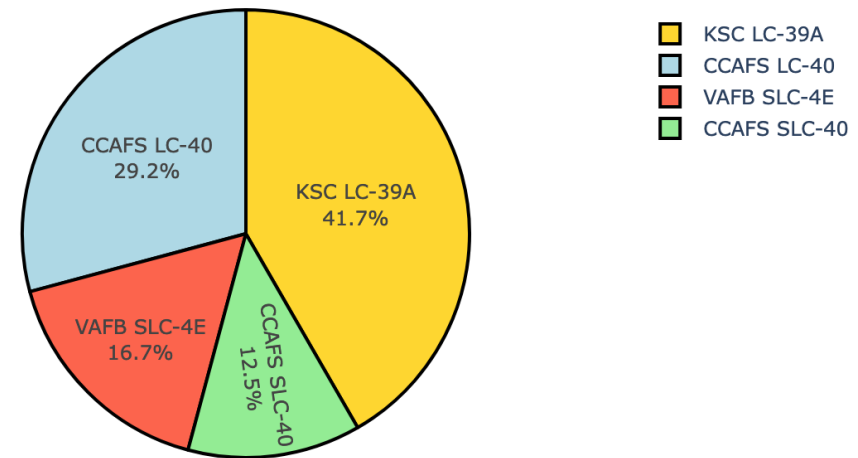
- KSC LC-39A experienced the highest proportion of successful landings, followed by CCAFS LC-40.
- VAFB SLC-4E and CCAFS SLC-40 the lowest.

SpaceX Launch Records Dashboard

All Sites



Total Successful Launches by Site



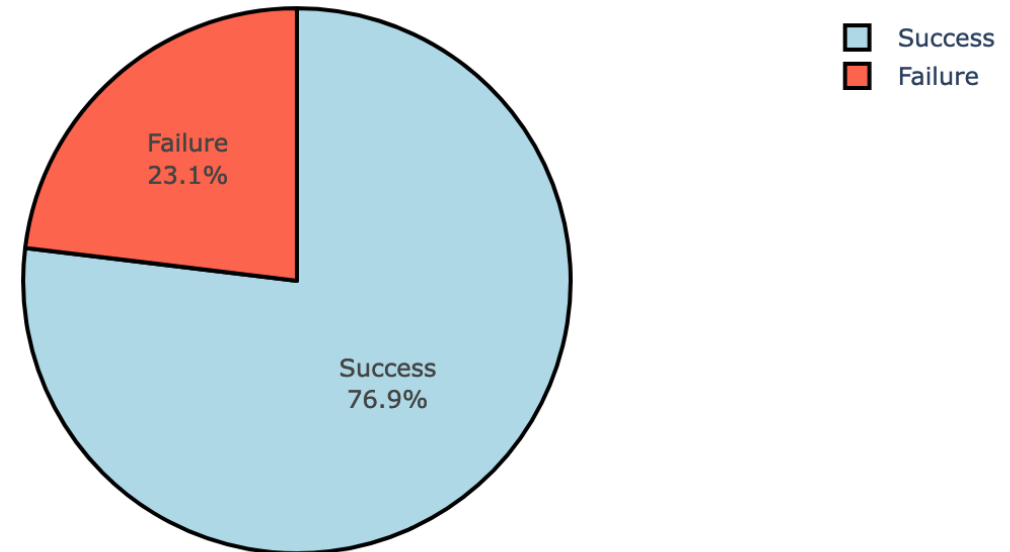
Per-site Launch Success Ratio: High

- KSC LC-39A had the highest ratio of successful landings

KSC LC-39A

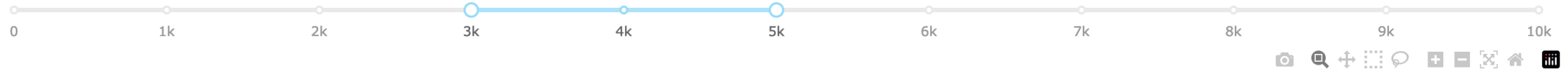


Launch Success vs Failure for site KSC LC-39A

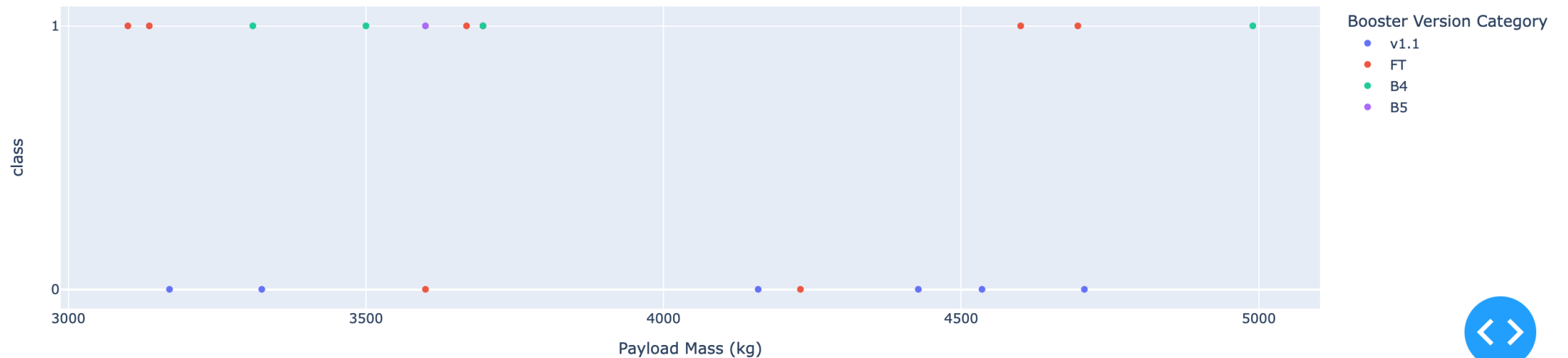


Payload Range

Payload range (Kg):



Payload Success Rate for All Sites

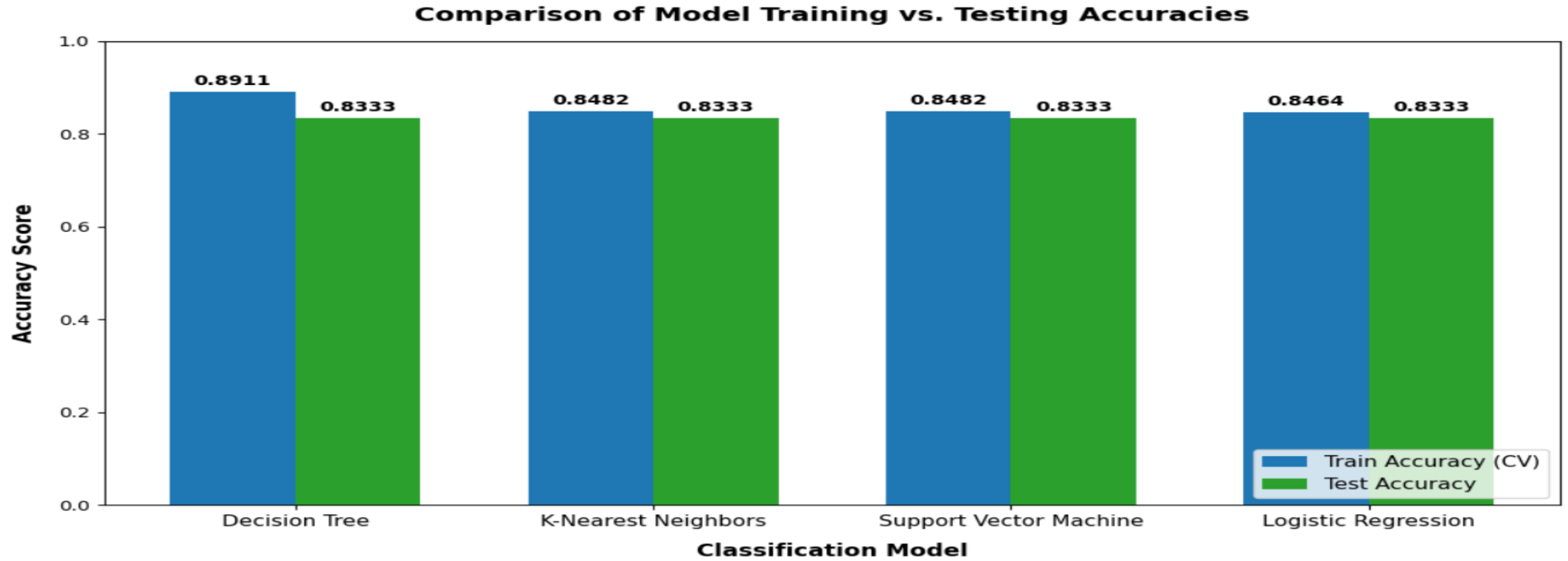


- With a payload mass between 3,000kg and 5,000kg, v1.1 boosters performed the worst.
- In the same payload range, B4 and B5 boosters had the best success rate, followed by FT.



Section 5

Predictive Analysis (Classification)

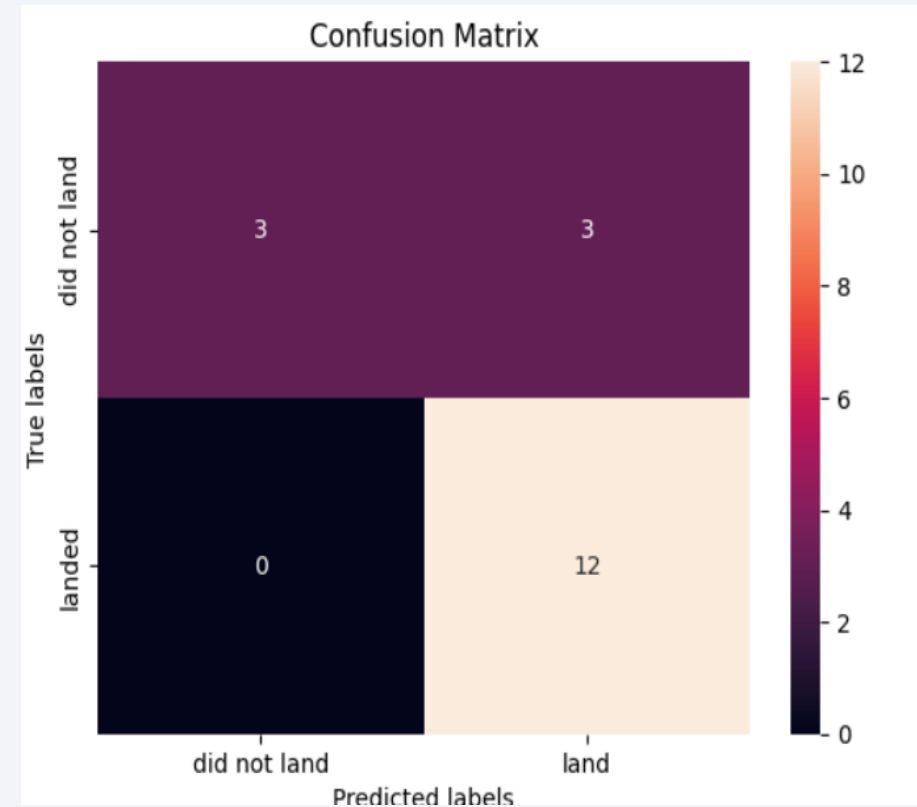


Classification Accuracy

Of the algorithms tested, the DecisionTreeClassifier was the most accurate per the Train and test.

Confusion Matrix

- 12 observations were correctly predicted as successful landings (true positive)
- 3 observations were correctly predicted as failed landings (true negative)
- 3 observation was incorrectly predicted as a successful landing (false positive)
- No observations were incorrectly predicted as a failed landing (false negative)



Conclusions

- Success rates increase over time, across all factors, which indicates continuous and incremental operational improvements and technological advancements.
- Different orbits have varying success rates, with ES-L1, SSO, HEO, and GEO showing consistently successful outcomes.
- Launch site was a highly predictive factor, with KSC LC-39A being a top performer, closely followed by CCAFS LC-40.
- Many of the predictive models evaluated were able to predict landing outcome with an acceptable level of accuracy. In the testing performed, DecisionTreeClassifier produced best results with high accuracy, precision, and recall.

Appendix

- Data Sources

- SpaceX API

- Collected Data: dataset_part_1.csv

- Wikipedia: List of Falcon 9 and Falcon Heavy launches (June 2021)

- Data after wrangling: dataset_part_2.csv

- Geographical data: spacex_launch_geo.csv

- Interactive data source: spacex_launch_dash.csv

Thank you!

